

Kristin Henderson

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Data Scientist | Applied ML, Statistical Modeling & GenAI

TECHNICAL SKILLS

Languages & Data: Python | R | SQL | SAS | MATLAB | Bash | PostgreSQL | MySQL | Parquet

Methods: Machine Learning | Statistical Modeling | Nonparametric Statistics | Time-Series Forecasting | Computer Vision

Generative AI & NLP: Generative AI | Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG) | Natural Language Processing (NLP) | LoRA Fine-Tuning

Frameworks & Libraries: pandas | scikit-learn | PyTorch | TensorFlow | XGBoost | Hugging Face | FAISS | OpenCV

Cloud & Tools: AWS (RDS, EC2) | Git | GitHub | Jupyter | Quarto | AWS Certified AI Practitioner

Visualization: matplotlib | seaborn | ggplot2 | R Shiny | D3.js | Tableau

PROFESSIONAL EXPERIENCE

SOUTHERN METHODIST UNIVERSITY, Dallas, TX

Jan 2026 – Apr 2026

Data Science Capstone (Client Project)

- Built a multi-criteria decision analysis framework in R, solo, for a confidential global consumer-packaged goods client, scoring the investment potential of about 22,500 brand extensions across a 77-million-row retail dataset.
- Validated against hand-labeled data at 6 to 18 months; median year-over-year case sales growth separated low-performing from viable brands with up to 85% balanced accuracy; ordinal logistic regression reached 93% adjacent accuracy.
- Built a 4-stage R pipeline ending in a self-contained interactive dashboard that stakeholders view locally; presented to the client and about 40 others, with a paper forthcoming in the SMU Data Science Review.

HOWARD HUGHES MEDICAL INSTITUTE, JANELIA RESEARCH CAMPUS, Ashburn, VA

Jun 2015 – Present

Research Technician III, Jayaraman Lab

Dec 2021 – Present

- Build semi-automated Python pipelines that process 100+ GB/day of behavioral video, applying statistical analysis and machine learning, including YOLO object detection and tracking, to characterize behavior.
- Directed design of the institute's first automated Danionella rig, with closed-loop feeders, 3 cameras, projected stimuli, and a hydrophone, built with engineers; co-developed its control software, which a colleague now uses.
- Benchmarked fish trackers against hand-labeled frames, scoring position error, detection rate, and identity swaps, and reduced swaps with Hungarian and greedy trajectory stitching; adding contrastive learning to a YOLO model, in progress.
- Applied nonparametric tests with multiple-comparison correction; all 9 fish learned a closed-loop task to some extent.
- Documented rig workflows; onboarded 2 lab members and trained 1 researcher to run the rig.

Research Technician II, Keleman Lab

Jan 2017 – Dec 2021

- Second author on a 2022 Nature Communications paper on learning and sleep in Drosophila long-term memory; ran the experiments and analyzed the data for four figure panels covering about 900 flies, using permutation and Wilcoxon tests.
- Took over an unpublished dopamine neuron memory project, independently redoing all experiments, analysis, and figures.

Research Technician I, Project Technical Resources

Jun 2015 – Dec 2016

- Processed 20 to 40 dissection samples per day for a large-scale fly neuron screening project, keeping low-quality samples from reaching the confocal microscope; co-authored the 2025 eLife paper on the split-GAL4 driver line resource.
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SELECTED PROJECTS

- **Custom RAG System (solo, 2025):** Trained two embedding models from scratch with contrastive learning, LoRA fine-tuned Gemma-3-1b-it, and built a FAISS retrieval pipeline with a cross-encoder reranker, benchmarked with Tonic Validate.
 - **Cost-Sensitive Classification (case study, Aug 2025):** Compared XGBoost and a PyTorch neural network on 160,000 records, tuning thresholds to minimize misclassification cost; the neural network cut total cost by just over half, at about 50x the training time.
 - **River Flow Forecasting (solo coursework, Dec 2025):** Forecasted river flow at 7-, 30-, and 365-day horizons in R, comparing ARMA, VAR, regression, neural network, and ensemble models; the best model differed by horizon.
 - **Stats Notes for Data Science (personal project, Aug 2025):** Authored a 25-chapter open-source statistics reference in Quarto with worked R and SAS examples, expanded from graduate course notes.
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EDUCATION

Master of Science in Data Science, May 2026, Southern Methodist University, Dallas, TX

- MSDS Outstanding Graduate Award

Bachelor of Science, emphasis in Biology, The Evergreen State College, Olympia, WA